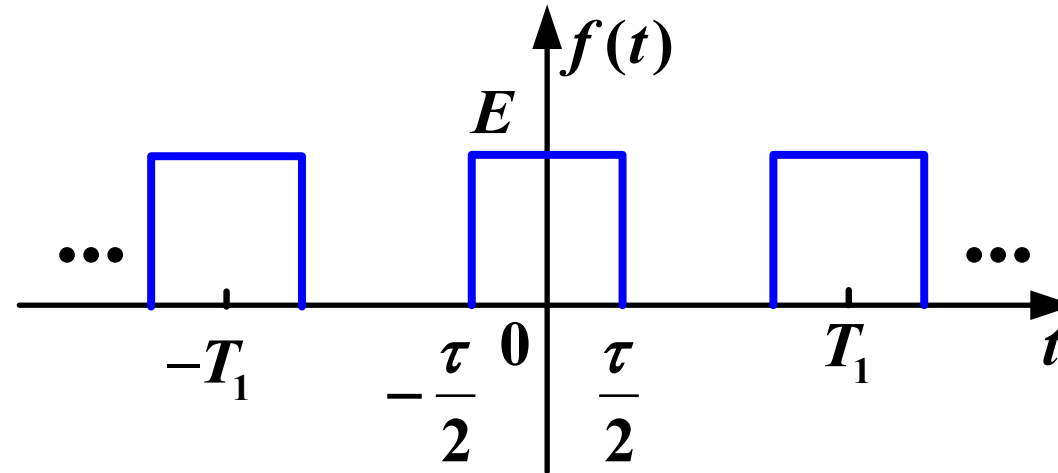


Main Content

- Derivation of Fourier Transform
- Inverse Fourier Transform
- Representations of Fourier Transform
- Physical Interpretation
- Existence Conditions

1. Derivation of Fourier Transform



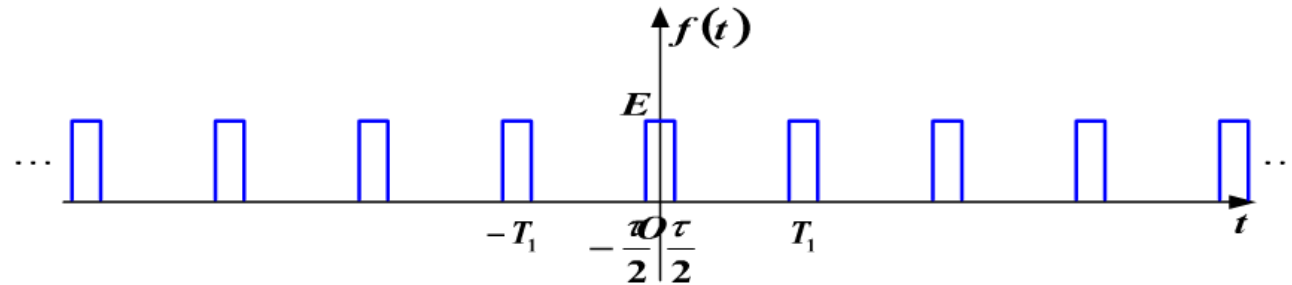
$$f(t) = \sum_{n=-\infty}^{\infty} F_k e^{j k \omega_1 t} \quad \left(\omega_1 = \frac{2\pi}{T_1} \right)$$

$$F_k = F(k \omega_1) = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-j k \omega_1 t} dt$$

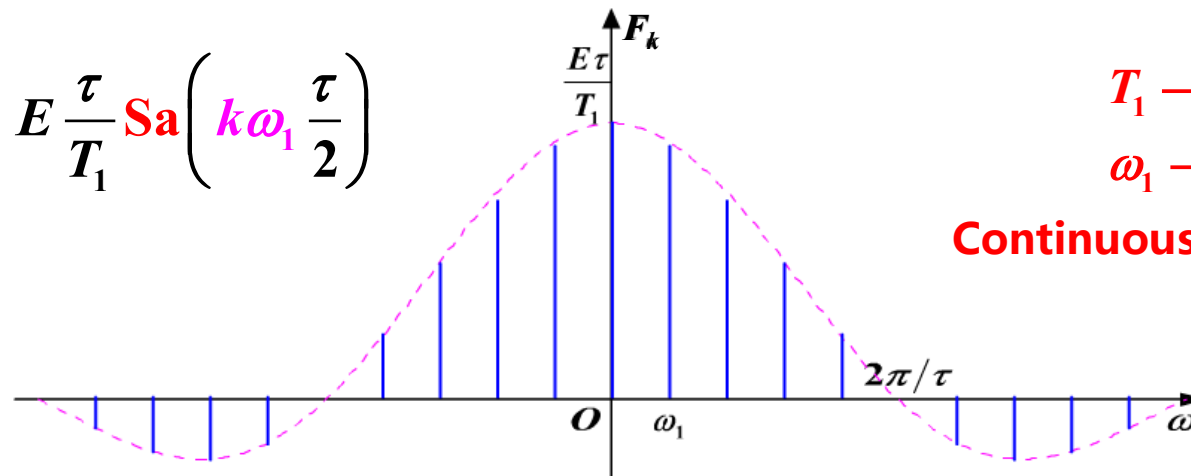
1. Derivation of Fourier Transform

Fourier series of a periodic rectangular pulse signal

τ remains unchanged, $T_1 \uparrow$



$$F_k = E \frac{\tau}{T_1} \text{Sa} \left(k \omega_1 \frac{\tau}{2} \right)$$



$T_1 \rightarrow \infty$

$\omega_1 \rightarrow 0$

Continuous spectrum

$$\omega_1 = 2\pi/T_1$$

1. Derivation of Fourier Transform

$$F(k\omega_1) = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jk\omega_1 t} dt$$

Multiply both sides by T_1 , let $T_1 \rightarrow \infty$

$$\lim_{T_1 \rightarrow \infty} T_1 F(k\omega_1) = \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jk\omega_1 t} dt$$

Spectral Density Function
Abbreviated as
Spectrum Function

$$= \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$
$$= F(\omega)$$

$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0$$

Discrete

$$k\omega_1 \rightarrow \omega$$

Continuous

2. Inverse Fourier Transform

From the complex exponential form of the Fourier series

$$\begin{aligned} f(t) &= \sum_{k=-\infty}^{\infty} F(k\omega_1) e^{jk\omega_1 t} \\ &= \sum_{k=-\infty}^{\infty} \frac{F(k\omega_1)}{\omega_1} \cdot e^{jk\omega_1 t} \cdot \omega_1 \\ &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} T_1 F(k\omega_1) \cdot e^{jk\omega_1 t} \cdot \omega_1 \end{aligned}$$

When $T_1 \rightarrow \infty$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot e^{j\omega t} d\omega$$

$$T_1 F(k\omega_1) \rightarrow F(\omega)$$

$$k\omega_1 \rightarrow \omega$$

$$\omega_1 \rightarrow d\omega$$

$$\sum_{k=-\infty}^{\infty} \rightarrow \int_{-\infty}^{\infty}$$

3. Representations of Fourier Transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \mathcal{F}[f(t)]$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega = \mathcal{F}^{-1}[F(\omega)]$$

$$f(t) \xleftrightarrow{\text{FT}} F(\omega)$$

Original function

Fourier transform function

3. Representations of Fourier Transform

$$F(\omega) = R(\omega) + jX(\omega)$$

$$F(\omega) = |F(\omega)| e^{j\varphi(\omega)}$$

$$|F(\omega)| \sim \omega \quad \text{Amplitude spectrum}$$

$$\varphi(\omega) \sim \omega \quad \text{Phase spectrum}$$

$|F(\omega)|$ Represents the **relative magnitudes** of various frequency components in the signal

$\varphi(\omega)$ Represents the **phase relationships** between the various frequency components of the signal

4. Physical Interpretation

$$\begin{aligned} f(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{F}(\omega) \mathbf{e}^{j\omega t} \mathbf{d}\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \mathbf{F}(\omega) \right| \mathbf{e}^{j\varphi(\omega)} \mathbf{e}^{j\omega t} \mathbf{d}\omega \\ &= \int_{-\infty}^{\infty} \left\{ \frac{\left| \mathbf{F}(\omega) \right|}{2\pi} \mathbf{e}^{j[\omega t + \varphi(\omega)]} \mathbf{d}\omega \right\} \end{aligned}$$

An infinite sum of continuous exponential signals with infinitesimal amplitudes

Frequency (ω) range: $-\infty \rightarrow \infty$

5. Existence Conditions

The Dirichlet conditions (sufficient conditions)

1. **Absolute Integrability**, $\int_{-\infty}^{\infty} |f(t)| dt < \infty$, ensures the Fourier integral converges
2. **Finite Extrema**, $f(t)$ must have a finite number of maxima and minima in any finite interval
3. **Finite Discontinuities**, $f(t)$ can have only a finite number of discontinuities in any finite interval. At each discontinuity, the function value must be finite (no infinite jumps)

Additional Notes:

- **Energy-Limited Signals**: The Fourier transform exists for signals with **finite energy**
- **Generalized Functions**: By introducing the Dirac delta function $\delta(\omega)$, the scope of functions admitting a Fourier transform expands significantly (e.g., constants, periodic signals)

Summary

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

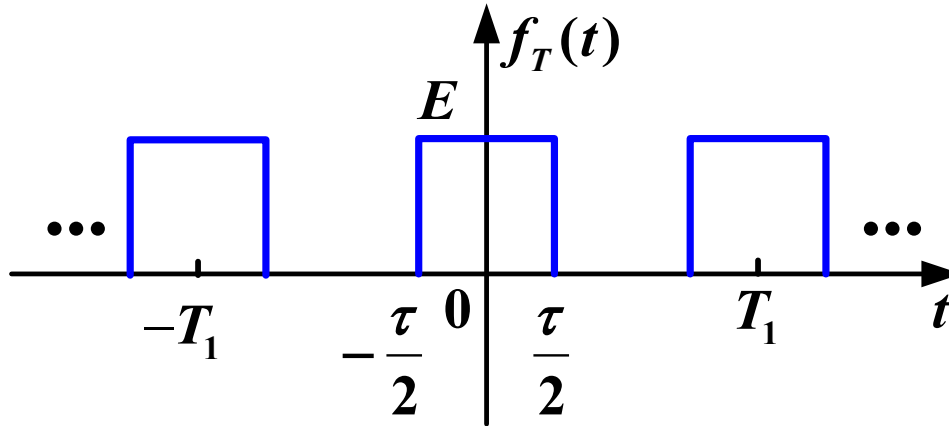
$$= \int_{-\infty}^{\infty} \left\{ \frac{|F(\omega)|}{2\pi} e^{j[\omega t + \phi(\omega)]} d\omega \right\}$$

Decomposing $f(t)$ into an infinite sum of continuous exponential signals with infinitesimal amplitudes

Main Content

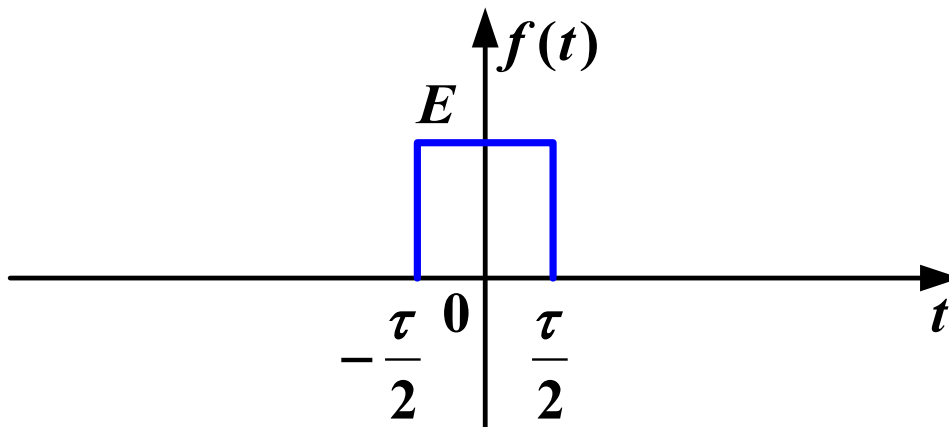
- Rectangular Pulse Signal
- One-sided Exponential Signal
- DC (Direct Current) Signal
- Signum Function (Sign Function)
- Unit Impulse Signal (Delta Function)
- Doublet Impulse Signal (Impulse Pair)

1. Rectangular Pulse Signal



$$F(k\omega_1) = \frac{E\tau}{T_1} \text{Sa}\left(\frac{k\omega_1\tau}{2}\right)$$

Introduction to Fourier transform
Multiply both sides by T_1



$$F(\omega) = \lim_{T_1 \rightarrow \infty} T_1 F(k\omega_1)$$

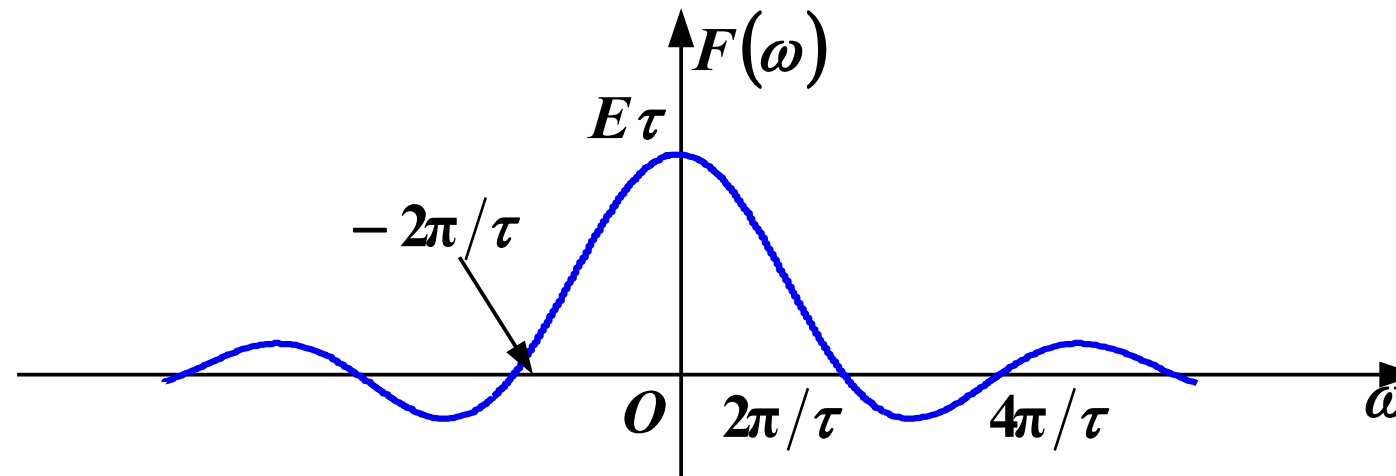
$$= E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

1. Rectangular Pulse Signal

$$F(\omega) = E\tau \operatorname{Sa}\left(\frac{\tau}{2} \cdot \omega\right)$$

$$\frac{\sin\left(\frac{\tau}{2}\omega\right)}{\frac{\tau}{2}\omega}$$

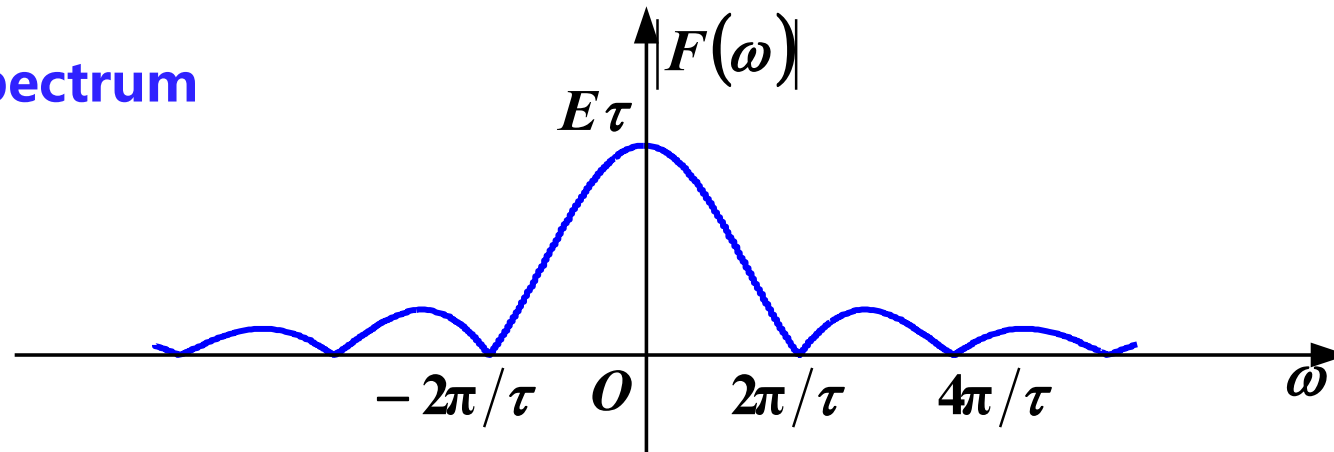
$$\frac{\tau}{2}\omega = K\pi$$



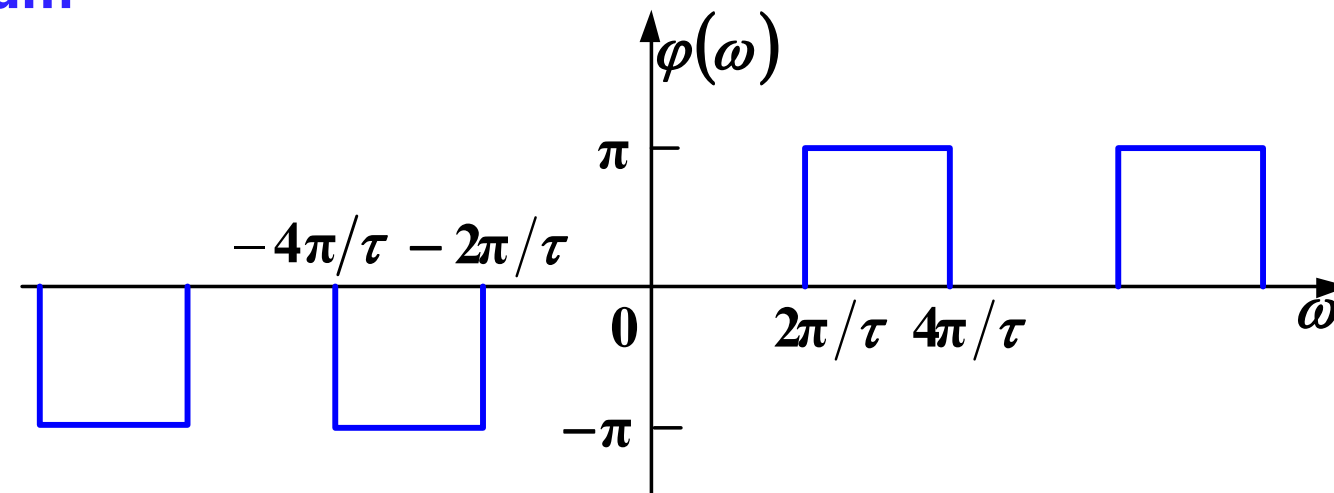
Bandwidth: $\frac{2\pi}{\tau}$

1. Rectangular Pulse Signal

Amplitude spectrum



Phase spectrum



2. One-sided Exponential Signal

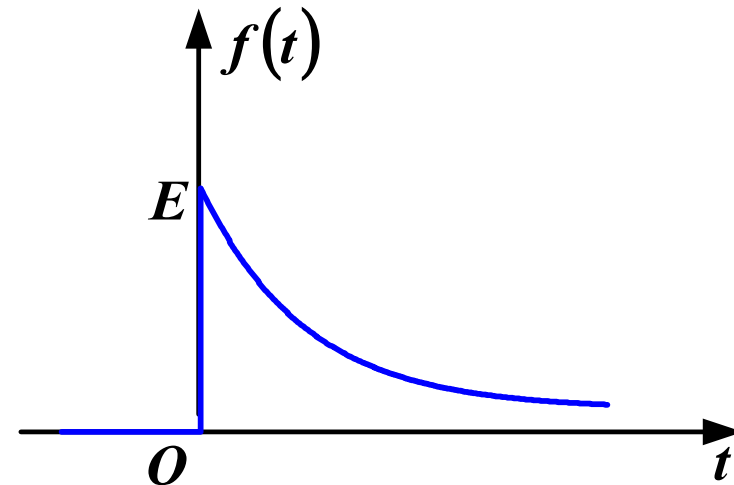
$$f(t) = E e^{-\alpha t} u(t) \quad (\alpha > 0)$$

$$F(\omega) = \int_{-\infty}^{\infty} E e^{-\alpha t} u(t) e^{-j\omega t} dt$$

$$= \int_0^{\infty} E e^{-(\alpha + j\omega)t} dt$$

$$= \frac{E}{-(\alpha + j\omega)} e^{-(\alpha + j\omega)t} \bigg|_0^{\infty}$$

$$= \frac{E}{\alpha + j\omega}$$

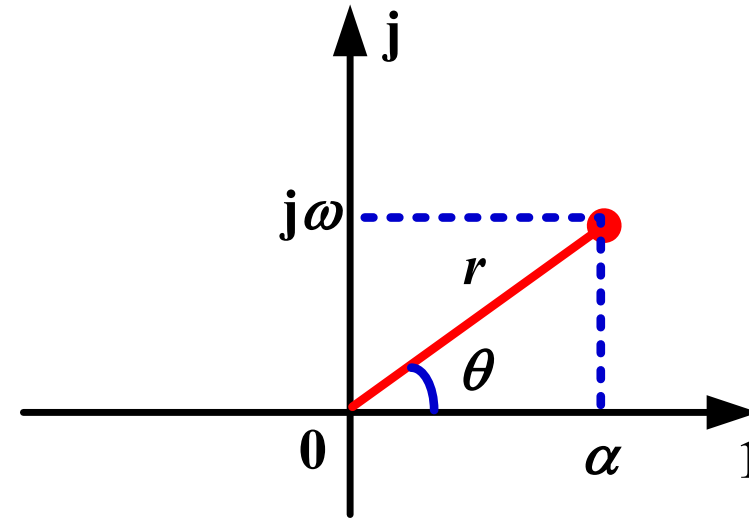


2. One-sided Exponential Signal

$$F(\omega) = \frac{E}{\alpha + j\omega} = \frac{E e^{j0}}{r e^{j\theta}}$$

$$|F(\omega)| = \frac{E}{\sqrt{\alpha^2 + \omega^2}}$$

$$\varphi(\omega) = -\arctan \frac{\omega}{\alpha}$$



$$r = \sqrt{\alpha^2 + \omega^2}$$

$$\tan \theta = \frac{\omega}{\alpha}$$

$$\theta = \arctan \frac{\omega}{\alpha} \quad (\alpha > 0)$$

2. One-sided Exponential Signal

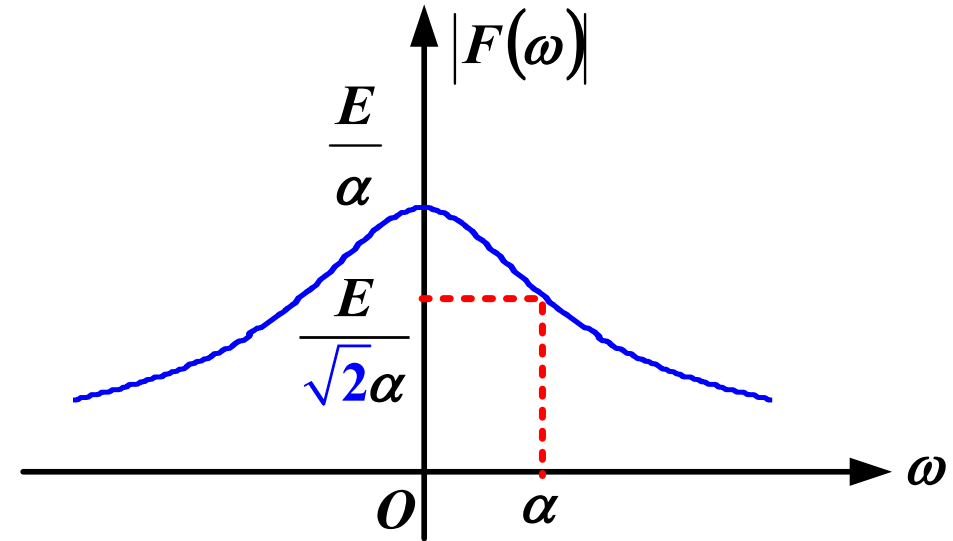
Amplitude spectrum

$$|F(\omega)| = \frac{E}{\sqrt{\alpha^2 + \omega^2}}$$

$$\bullet \omega = 0 \quad |F(\omega)| = \frac{E}{\alpha}$$

$$\bullet \omega \rightarrow \pm\infty \quad |F(\omega)| \rightarrow 0$$

$$\bullet \omega = \alpha \quad |F(\omega)| = \frac{E}{\sqrt{2}\alpha}$$

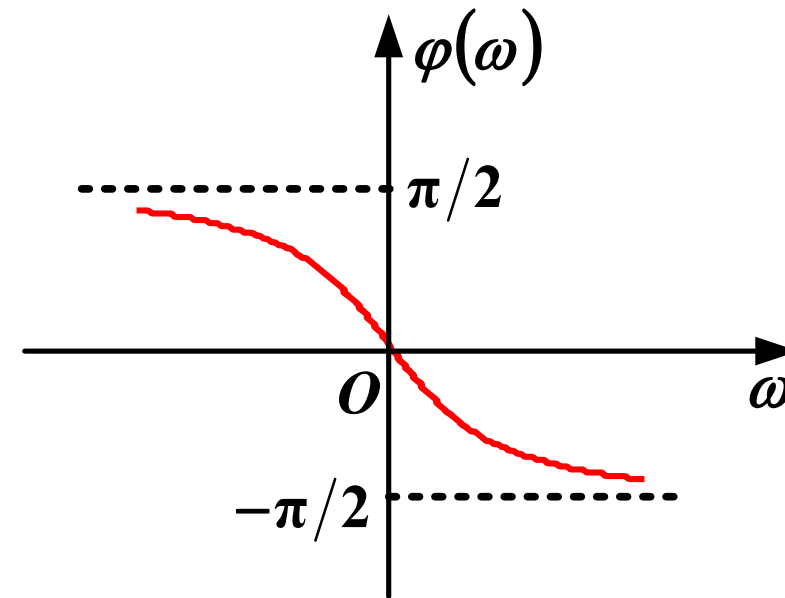


2. One-sided Exponential Signal

Phase spectrum

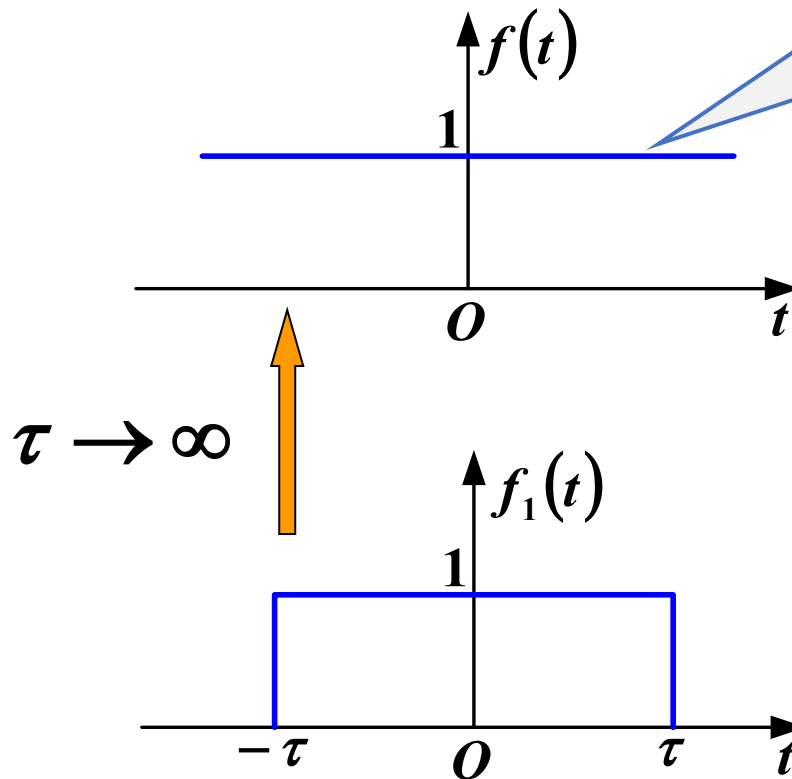
$$\varphi(\omega) = -\arctan \frac{\omega}{\alpha}$$

$$\begin{cases} \omega \rightarrow 0, & \varphi(\omega) = 0 \\ \omega \rightarrow +\infty, & \varphi(\omega) \rightarrow -\pi/2 \\ \omega \rightarrow -\infty, & \varphi(\omega) \rightarrow \pi/2 \end{cases}$$



3. DC (Direct Current) Signal

$$f(t) = 1, \quad -\infty < t < +\infty$$



Does not satisfy the absolute integrability condition and cannot be directly evaluated using the definition

$$F_1(\omega) = 2\tau \text{Sa}(\omega\tau)$$

3. DC (Direct Current) Signal

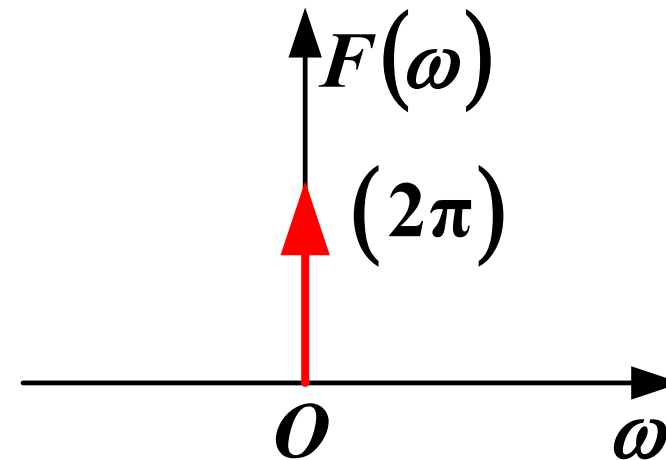
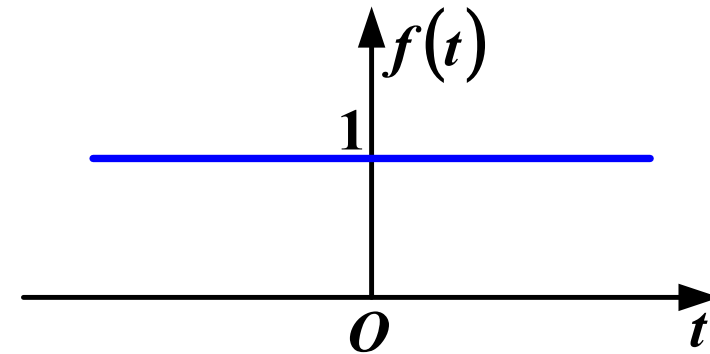
$$F(\omega) = \lim_{\tau \rightarrow \infty} F_1(\omega)$$

$$= 2\pi \lim_{\tau \rightarrow \infty} \frac{\tau}{\pi} \text{Sa}(\omega\tau)$$

$$= 2\pi \delta(\omega)$$

$$= 2\pi \delta(2\pi f)$$

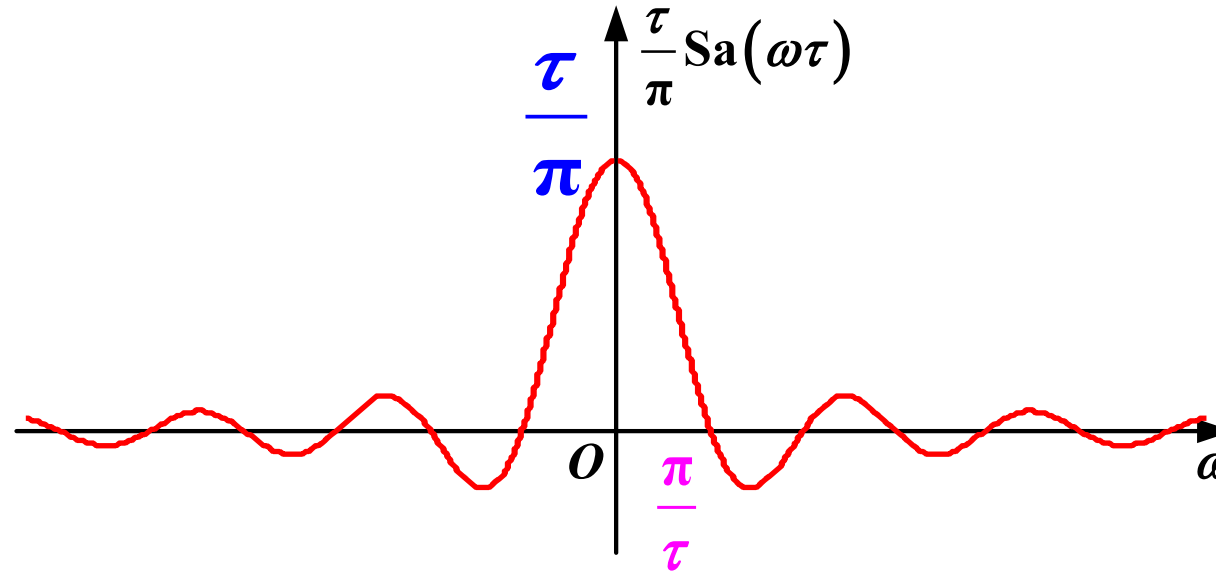
$$= \delta(f)$$



$$\delta(at) = \frac{1}{|a|} \delta(t)$$

3. DC (Direct Current) Signal

Explanation



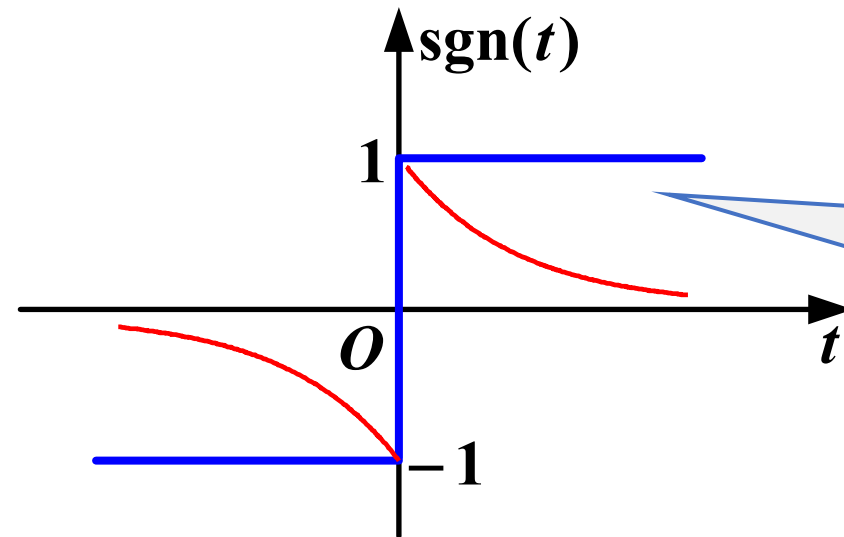
$\tau \uparrow, \frac{\pi}{\tau} \downarrow$ waveform compression, $\frac{\tau}{\pi} \uparrow, \omega=0$ point value increases

$\tau \rightarrow \infty$, energy compressed to $\omega = 0$, area $\frac{\tau}{\pi} \frac{\pi}{\tau} = 1$ always the same

$$\lim_{\tau \rightarrow \infty} \frac{\tau}{\pi} \text{Sa}(\omega\tau) = \delta(\omega)$$

4. Signum Function (Sign Function)

$$f(t) = \text{sgn}(t)$$



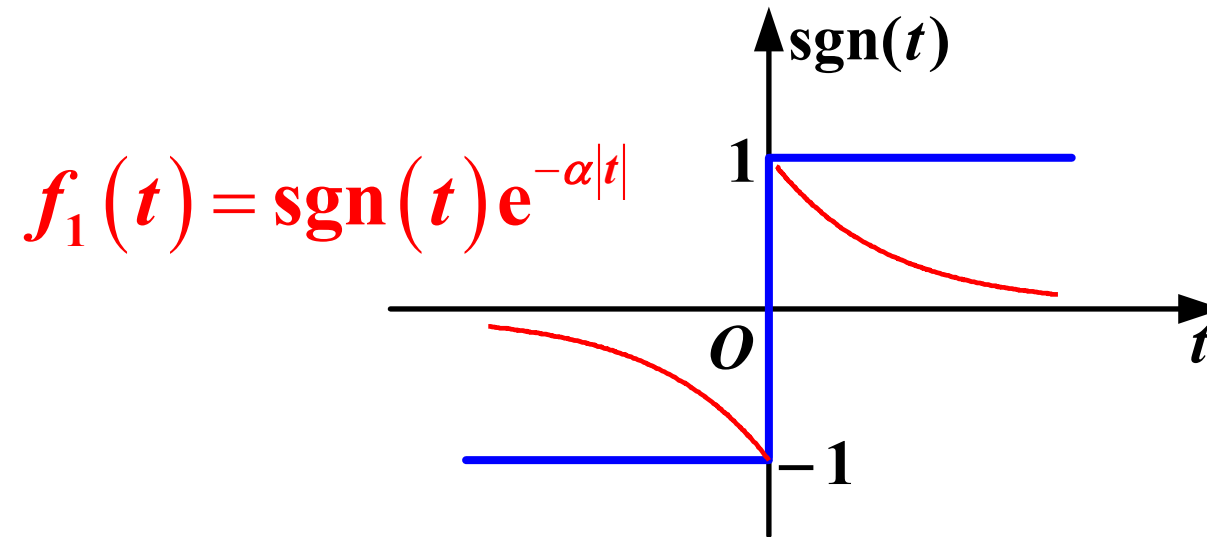
does not satisfy
the absolute
integrability
condition

Processing method: constructing a two-sided exponential function

$$f_1(t) = \text{sgn}(t) e^{-\alpha|t|}$$

$$F(\omega) = \lim_{\alpha \rightarrow 0} F_1(\omega)$$

4. Signum Function (Sign Function)



$$\begin{aligned} F_1(\omega) &= \int_{-\infty}^0 -e^{\alpha t} e^{-j\omega t} dt + \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt \\ &= \frac{-1}{\alpha - j\omega} + \frac{1}{\alpha + j\omega} \\ &= \frac{-j2\omega}{\alpha^2 + \omega^2} \end{aligned}$$

4. Signum Function (Sign Function)

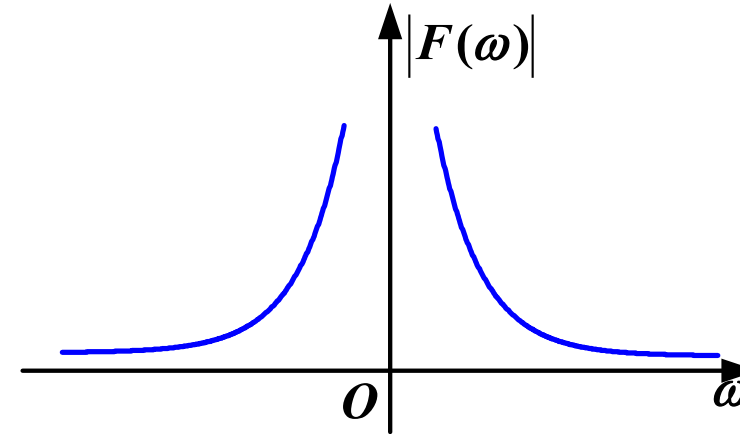
$$\begin{aligned} F(\omega) &= \lim_{\alpha \rightarrow 0} F_1(\omega) \\ &= \lim_{\alpha \rightarrow 0} \frac{-j2\omega}{\alpha^2 + \omega^2} \\ &= \frac{2}{j\omega} \\ &= \begin{cases} \frac{2}{|\omega|} e^{-j\frac{\pi}{2}} & \omega > 0 \\ \frac{2}{|\omega|} e^{j\frac{\pi}{2}} & \omega < 0 \end{cases} \end{aligned}$$

4. Signum Function (Sign Function)

$$F(\omega) = \lim_{\alpha \rightarrow 0} \frac{-j2\omega}{\alpha^2 + \omega^2} = \frac{2}{j\omega}$$

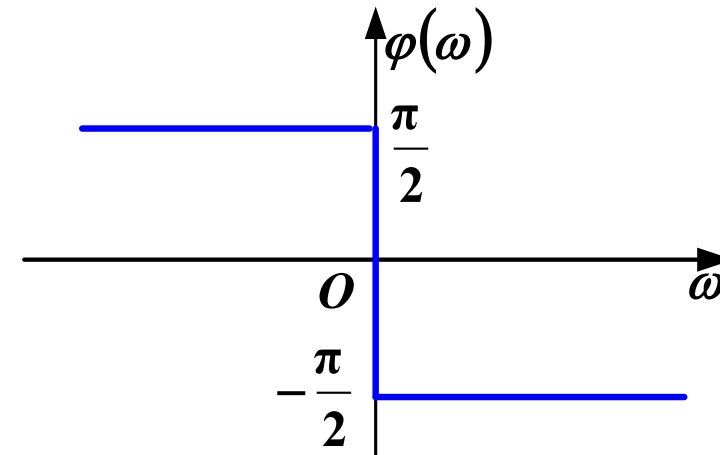
Amplitude spectrum

$$|F(\omega)| = \frac{2}{|\omega|}$$



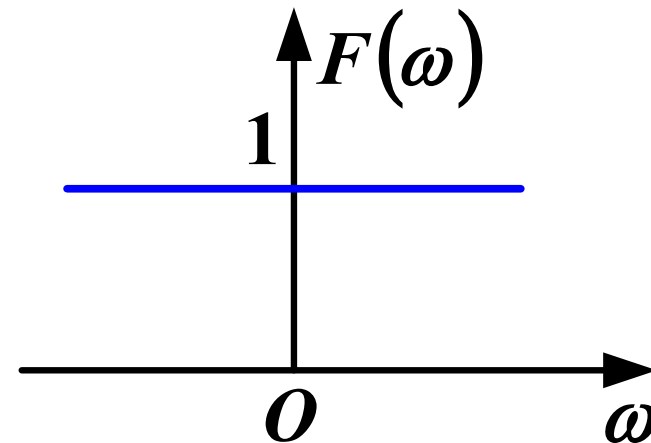
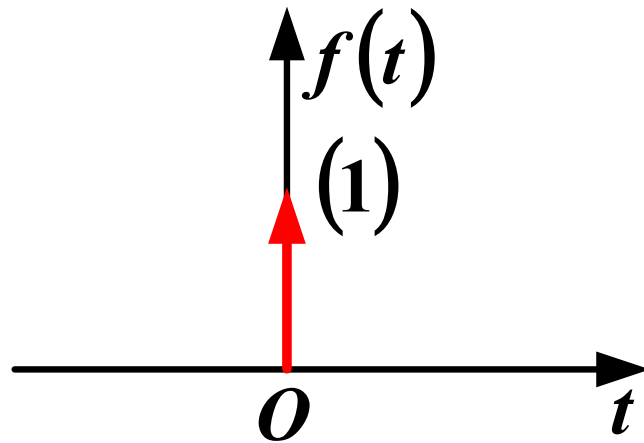
Phase spectrum

$$\begin{cases} -\frac{\pi}{2}, & \omega > 0 \\ \frac{\pi}{2}, & \omega < 0 \end{cases}$$



5. Unit Impulse Signal (Delta Function)

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$



**Infinitesimally narrow in the time domain,
infinitely wide in the frequency domain**

6. Doublet Impulse Signal (Impulse Pair)

$$\begin{aligned}\mathcal{F}[\delta'(t)] &= \int_{-\infty}^{\infty} \delta'(t) e^{-j\omega t} dt \\ &= -\left(-j\omega e^{-j\omega t}\right)\Big|_{t=0} \\ &= j\omega\end{aligned}$$

$$\begin{aligned}\int_{-\infty}^{\infty} f(t) \delta'(t) dt \\ &= -f'(0)\end{aligned}$$

Summary

- Rectangular Pulse Signal
- One-sided Exponential Signal
- DC (Direct Current) Signal
- Signum Function (Sign Function)
- Unit Impulse Signal (Delta Function)
- Doublet Impulse Signal (Impulse Pair)

Main Content

The Fourier transform is unique. The properties of the Fourier transform reveal the deterministic **intrinsic relationship** between the time-domain and frequency-domain characteristics of signals. The purpose of discussing the properties of the Fourier transform is to:

- Understand the intrinsic connections of these properties
- Utilize the properties to derive the Fourier transform of certain signals
- Understand the applications of the Fourier transform in communication systems

1. Linearity

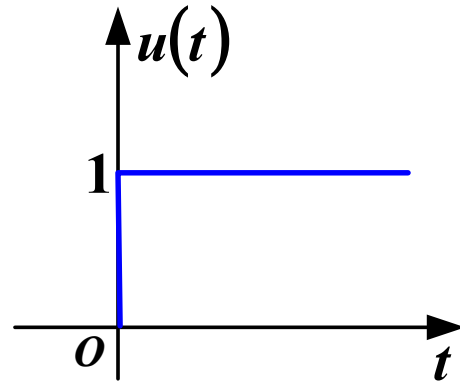
If $\mathcal{F}[f_i(t)] = F_i(\omega)$ ($i = 1, 2, \dots, n$), Then

$$\mathcal{F}\left[\sum_{i=1}^n a_i f_i(t)\right] = \sum_{i=1}^n a_i F_i(\omega)$$

a_i are constants, n is a positive integer

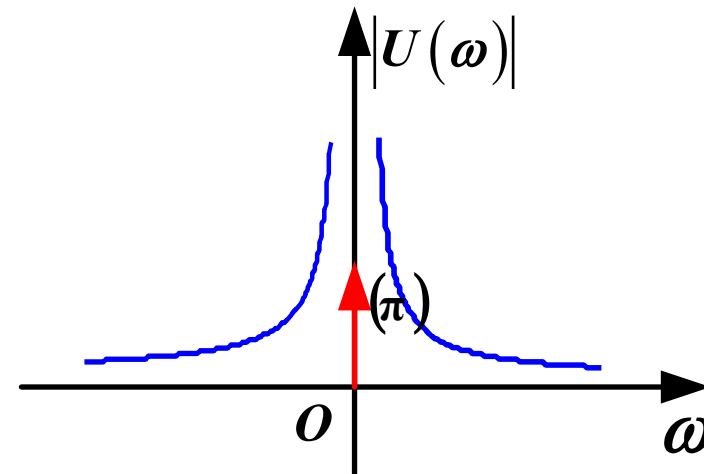
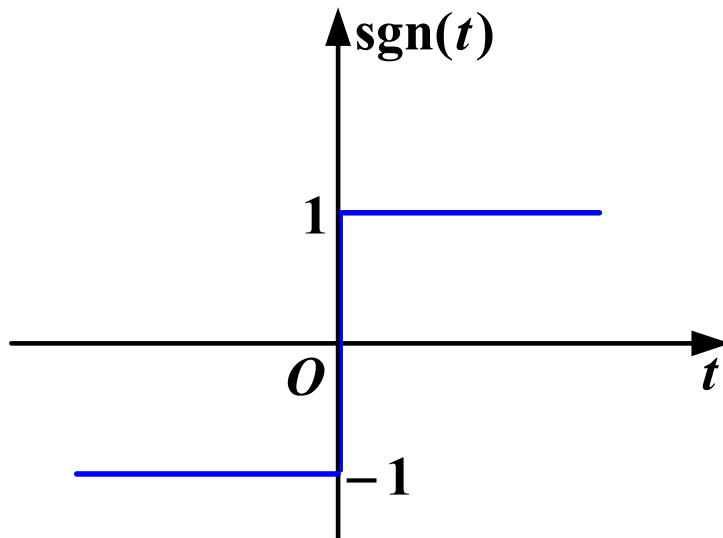
6. Basic Properties of Fourier Transform

Example 1 Fourier transform of a unit step signal



$$u(t) = \frac{1}{2} + \frac{1}{2} \operatorname{sgn}(t)$$

$$\xleftrightarrow{\text{FT}} \pi \delta(\omega) + \frac{1}{j\omega}$$



2. Duality

(1) Statement

$$\text{If } \mathcal{F}[f(t)] = F(\omega), \text{ Then } \mathcal{F}[F(t)] = 2\pi f(-\omega)$$

(2) Analysis

$$\begin{cases} \text{FT: } F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ \text{IFT: } f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \end{cases}$$

The forward and inverse transforms are structurally similar but not identical. The duality property formalizes this symmetry

6. Basic Properties of Fourier Transform

Example 2

Given $f(t) = \delta(t)$, $F(\omega) = 1$

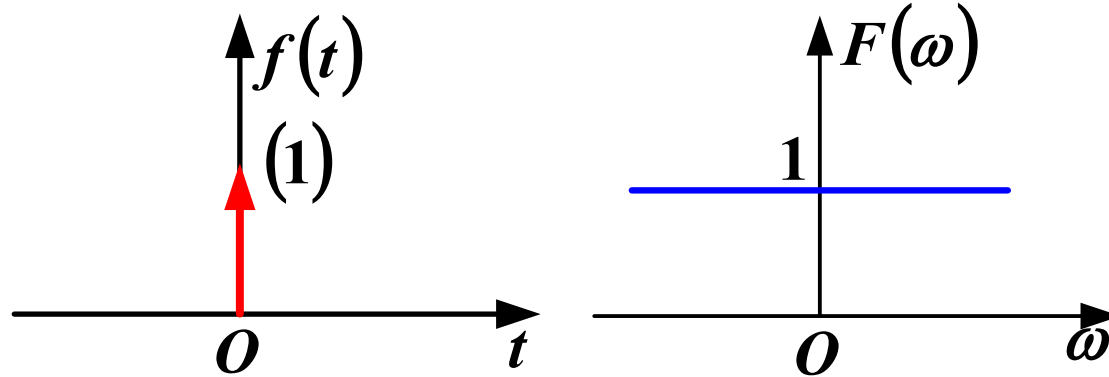
Use the duality property to find the Fourier transform of $F(t) = 1$

$$\begin{aligned}\mathcal{F}[F(t)] &= 2\pi f(-\omega) \\ &= 2\pi\delta(-\omega) \quad (\delta(\omega) \text{ is even function}) \\ &= 2\pi\delta(\omega)\end{aligned}$$

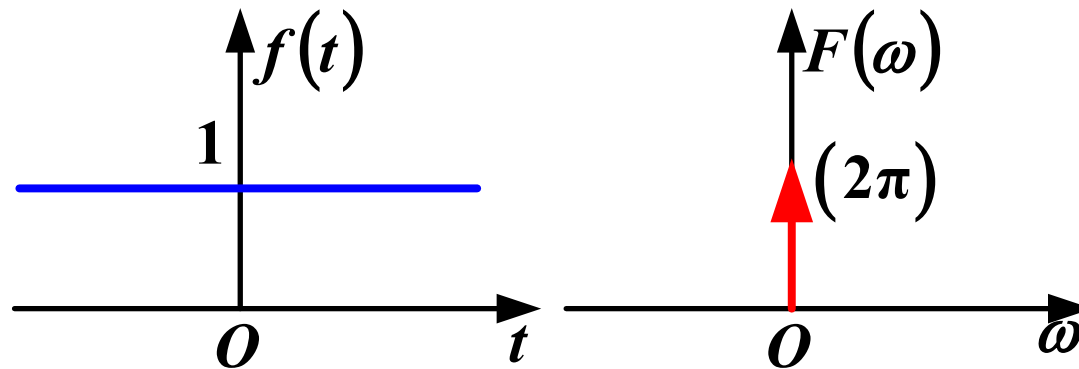
6. Basic Properties of Fourier Transform

Example 2

$$\delta(t) \leftrightarrow 1$$



$$1 \leftrightarrow 2\pi\delta(\omega)$$



Duality Property

6. Basic Properties of Fourier Transform

Example 4

Find the Fourier transform of the sampling signal $\text{Sa}(\omega_c t)$

Analysis

Using the Fourier transform definition:

$$F_1(\omega) = \int_{-\infty}^{\infty} \text{Sa}(\omega_c t) e^{-j\omega t} dt$$

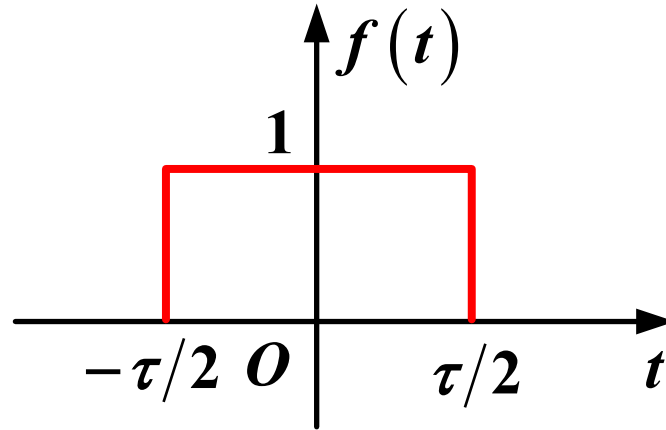
$$\int_{-\infty}^{\infty} \text{Sa}(\omega_c t) e^{-j\omega t} dt$$

This integral is challenging to solve directly

Therefore, we attempt to solve it using **duality property**

6. Basic Properties of Fourier Transform

Example 4

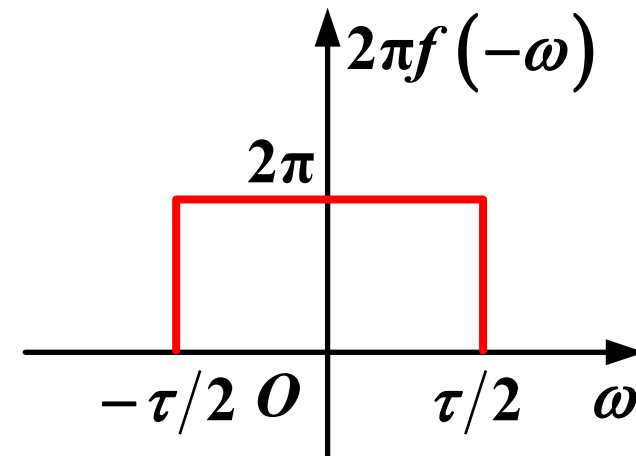


$$\Leftrightarrow F(\omega) = \tau \text{Sa}\left(\frac{\tau}{2}\omega\right)$$

Obtain a new
transform pair using
duality property

$$F(t) = \tau \text{Sa}\left(\frac{\tau}{2}t\right)$$

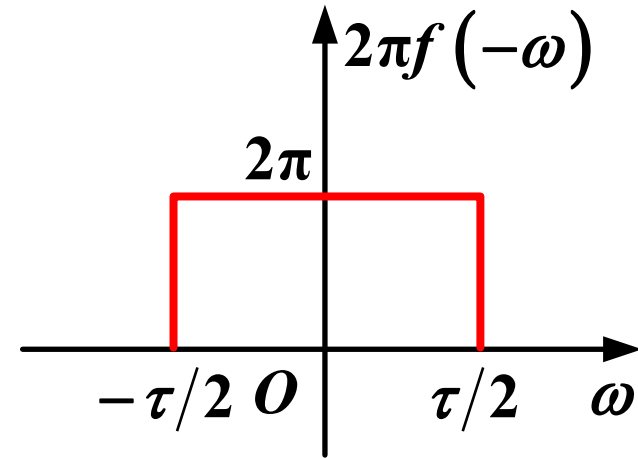
$\Leftrightarrow$



6. Basic Properties of Fourier Transform

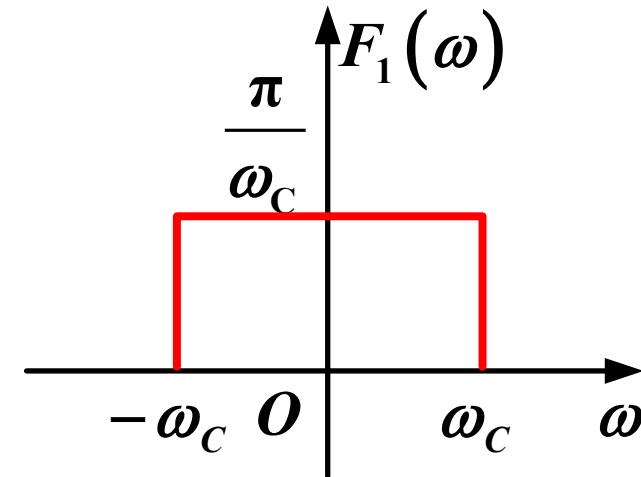
Example 4

$$F(t) = \tau \text{Sa}\left(\frac{\tau}{2}t\right) \quad \leftrightarrow$$



Let $\frac{\tau}{2} = \omega_c$, divide both sides by $2\omega_c$

$$\text{Sa}(\omega_c t) \quad \leftrightarrow$$



3. Conjugation Symmetry

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F(\omega) = |F(\omega)| e^{j\phi(\omega)} = R(\omega) + jX(\omega)$$

Magnitude **Phase** **Real part** **Imaginary part**

6. Basic Properties of Fourier Transform

(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions

$$\underbrace{f(t)}_{\text{Real signal}} = \underbrace{f_e(t)}_{\text{Even}} + \underbrace{f_o(t)}_{\text{Odd}}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Euler formula

$$= \int_{-\infty}^{\infty} [f_e(t) + f_o(t)] \cdot [\cos(\omega t) - j \sin(\omega t)] dt$$

$$= 2 \int_0^{\infty} f_e(t) \cos(\omega t) dt - j 2 \int_0^{\infty} f_o(t) \sin(\omega t) dt$$

Real part

Imaginary part

$$= R(\omega) + j X(\omega)$$

The integral of an odd function is 0

6. Basic Properties of Fourier Transform

(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions

$$R(\omega) = 2 \int_0^{\infty} f_e(t) \cos(\omega t) dt$$

Even function of ω

$$X(\omega) = -2 \int_0^{\infty} f_o(t) \sin(\omega t) dt$$

Odd function of ω

$$|F(\omega)| = \sqrt{[R(\omega)]^2 + [X(\omega)]^2}$$

Even function of ω

$$\varphi(\omega) = \text{atan2}[X(\omega), R(\omega)]$$

Odd function of ω

$$F(-\omega) = R(-\omega) + jX(-\omega) = R(\omega) - jX(\omega)$$

$$= F^*(\omega)$$

Conjugate symmetry

Four-quadrant arctangent

6. Basic Properties of Fourier Transform

(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions

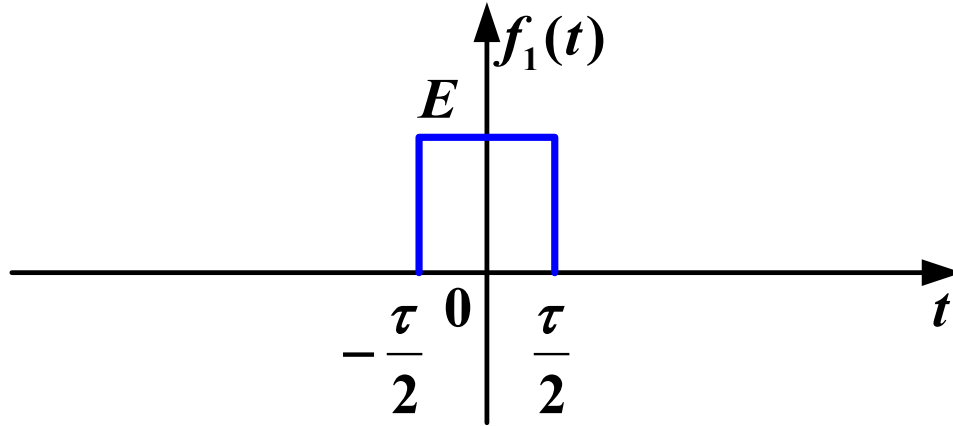
$$R(\omega) = 2 \int_0^{\infty} f_e(t) \cos(\omega t) dt \quad \text{Even function of } \omega$$

$$X(\omega) = -2 \int_0^{\infty} f_o(t) \sin(\omega t) dt \quad \text{Odd function of } \omega$$

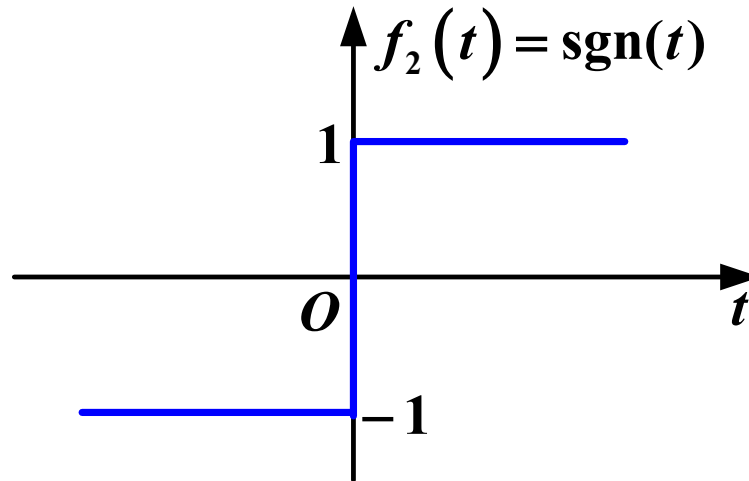
$f(t)$	$F(\omega) = \mathcal{F}[f(t)]$	
Even function: odd component is zero	Real even function: phase is 0 or π	$F(\omega) = R(\omega)$
Odd function: even component is zero	Imaginary odd function: phase is $\pm\pi/2$	$F(\omega) = \mathbf{j} X(\omega)$

6. Basic Properties of Fourier Transform

(1) Symmetry of the Spectrum for Real-Valued Time-Domain Functions



$$F_1(\omega) = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$



$$F_2(\omega) = \frac{2}{j\omega}$$

(2) Spectral Properties of Complex Time-Domain Signals

The Fourier transform of $f^*(t)$

$$\int_{-\infty}^{\infty} f^*(t) e^{-j\omega t} dt \quad (1)$$

Taking the complex conjugate twice on (1)

$$\left[\int_{-\infty}^{\infty} f(t) e^{j\omega t} dt \right]^* \quad (2)$$

Comparing with the forward Fourier transform, (2) equals $F^*(-\omega)$

$$F[f^*(t)] = F^*(-\omega) \quad (3)$$

4. Time-Scaling Property

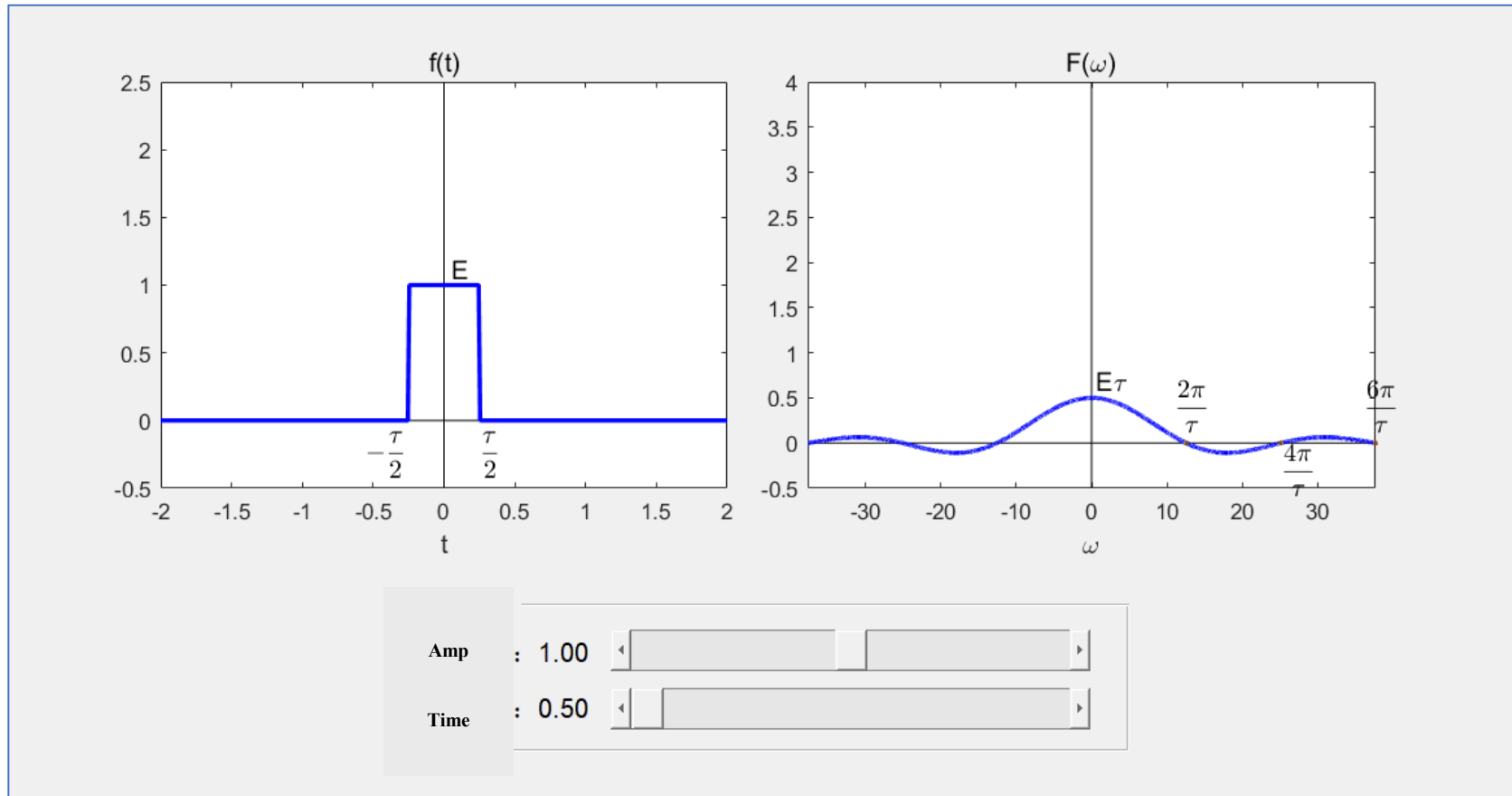
If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$,

Where a is a non-zero real constant

When $a = -1$ $f(t) \rightarrow f(-t)$, $F(\omega) \rightarrow F(-\omega)$

6. Basic Properties of Fourier Transform

4. Time-Scaling Property



Time duration and bandwidth are inversely proportional